Week 36: Making threes

Adult guide built from the problem files · Unpiloted. The K-1 readiness check, tile handling and timing are untested.

This week's adult guide is week-36-facilitator.pdf, whose source is editable/facilitator-src/guide.md. These problem files hold each problem's one-sentence idea, not its answers.

Problems and routes

Problem	K–1	2–3	4–5	route
Complete each pair	P1	P1	P1	first
Split the nine tiles into three allowed threes	P2	P2		first
The most tiles with no allowed three	P3	P3	P2	first
All the allowed threes through one tile	P4	P4	P3	return
Every split, and why the list is complete			P4	return
The adding game: can choices change its length?		P5		return
Can you get stuck early?			P5	return

First route. K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P4; Grades 2–3 P4–P5; Grades 4–5 P3–P5.

Key, one entry per problem

Complete each pair

K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Any two different tiles have exactly one completing tile: for each attribute keep the shared value or use the missing one.

Split the nine tiles into three allowed threes

K–1 P2 · Grades 2–3 P2 · first route · wording differs by band

Idea. The nine tiles split into three disjoint allowed threes in exactly four ways, the four parallel classes of the plane.

The most tiles with no allowed three

K–1 P3 · Grades 2–3 P3 · Grades 4–5 P2 · first route · wording differs by band

Idea. At most four tiles avoid an allowed three: five tiles have ten pairs, and the four unchosen tiles can complete at most two each.

All the allowed threes through one tile

K–1 P4 · Grades 2–3 P4 · Grades 4–5 P3 · return route · wording differs by band

Idea. Exactly four allowed threes pass through each tile, one for each other tile it pairs with, and two different threes share at most one tile.

Every split, and why the list is complete

Grades 4–5 P4 · return route

Idea. There are exactly four splits of the nine tiles into allowed threes, one for each direction of parallel lines.

The adding game: can choices change its length?

Grades 2–3 P5 · return route

Idea. Every legal collection of three tiles extends in exactly three ways and every stuck collection has four tiles, so the adding game always lasts four moves.

Can you get stuck early?

Grades 4–5 P5 · return route

Idea. Maximal equals maximum here: every three-tile legal collection has three legal fourth tiles, so no legal collection gets stuck before four tiles.